Name___________________________________

# The Law of Sines

Date_________________ Period____

**State the number of possible triangles that can be formed using the given measurements.**

1) $m\angle A = 31°$, $c = 20$ mi, $a = 16$ mi

2) $m\angle B = 82°$, $a = 34$ m, $b = 22$ m

3) $m\angle B = 110°$, $b = 11$ m, $a = 4$ m

4) $m\angle A = 64°$, $c = 33$ in, $a = 32$ in

**Find each measurement indicated. Round your answers to the nearest tenth.**

5) $m\angle A = 64°$, $m\angle B = 98°$, $a = 29$ mi
Find $b$

6) $m\angle A = 57°$, $c = 35$ cm, $a = 33$ cm
Find $b$

7) $m\angle C = 128°$, $b = 35$ in, $c = 35$ in
Find $a$

8) $m\angle C = 90°$, $m\angle B = 30°$, $b = 15$ in
Find $c$

9) In $\triangle TRS$, $m\angle S = 118°$, $s = 16$ ft, $r = 5$ ft
Find $m\angle R$

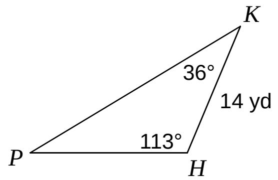
10) In $\triangle KHP$, $m\angle K = 27°$, $p = 35$ m, $k = 18$ m
Find $m\angle P$

11) In $\triangle RST$, $m\angle R = 153°$, $t = 29$ km, $r = 16$ km
Find $m\angle S$

12) In $\triangle KHP$, $m\angle K = 55°$, $p = 18$ m, $k = 27$ m
Find $m\angle P$

**Solve each triangle. Round your answers to the nearest tenth.**

13)

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